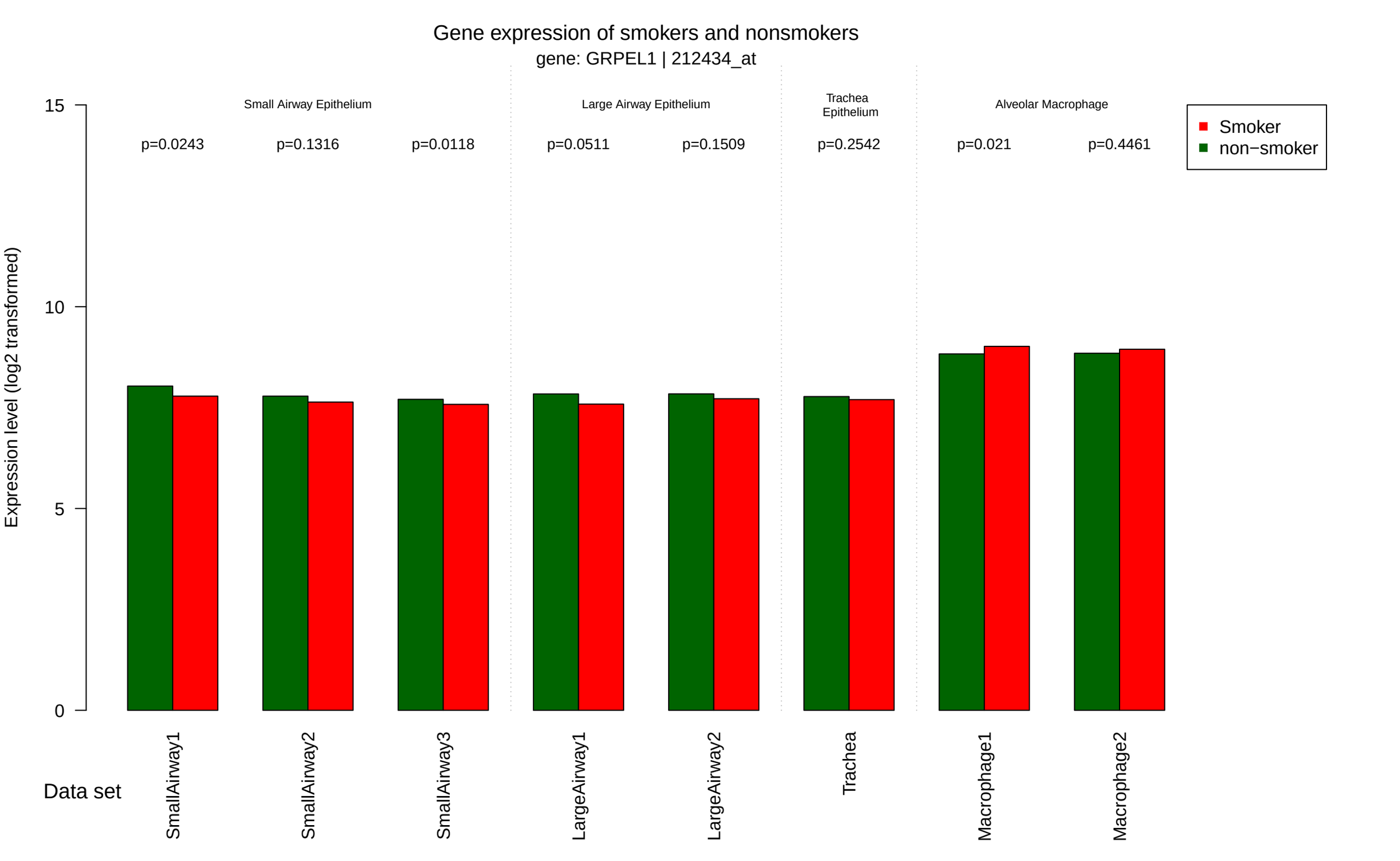
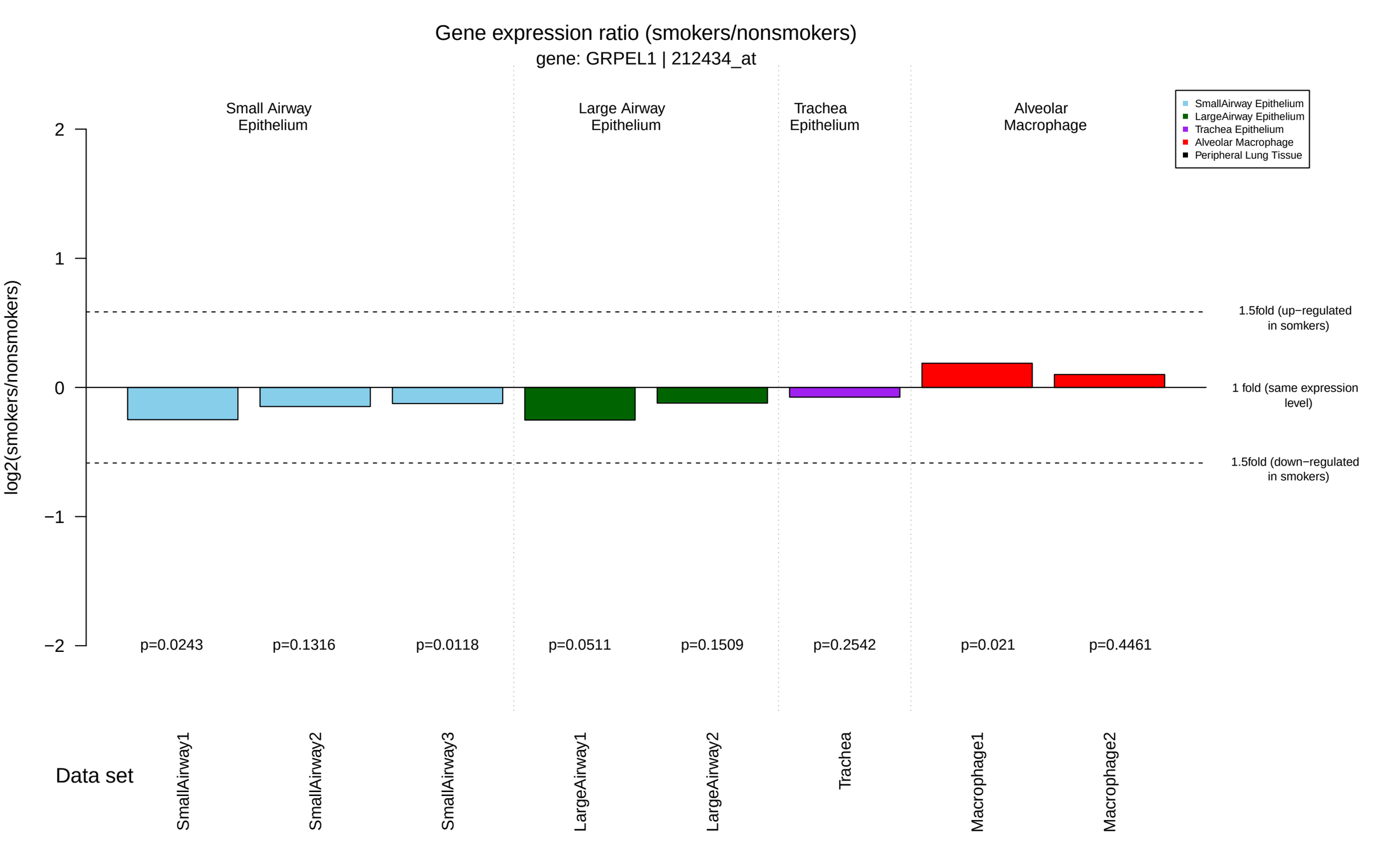
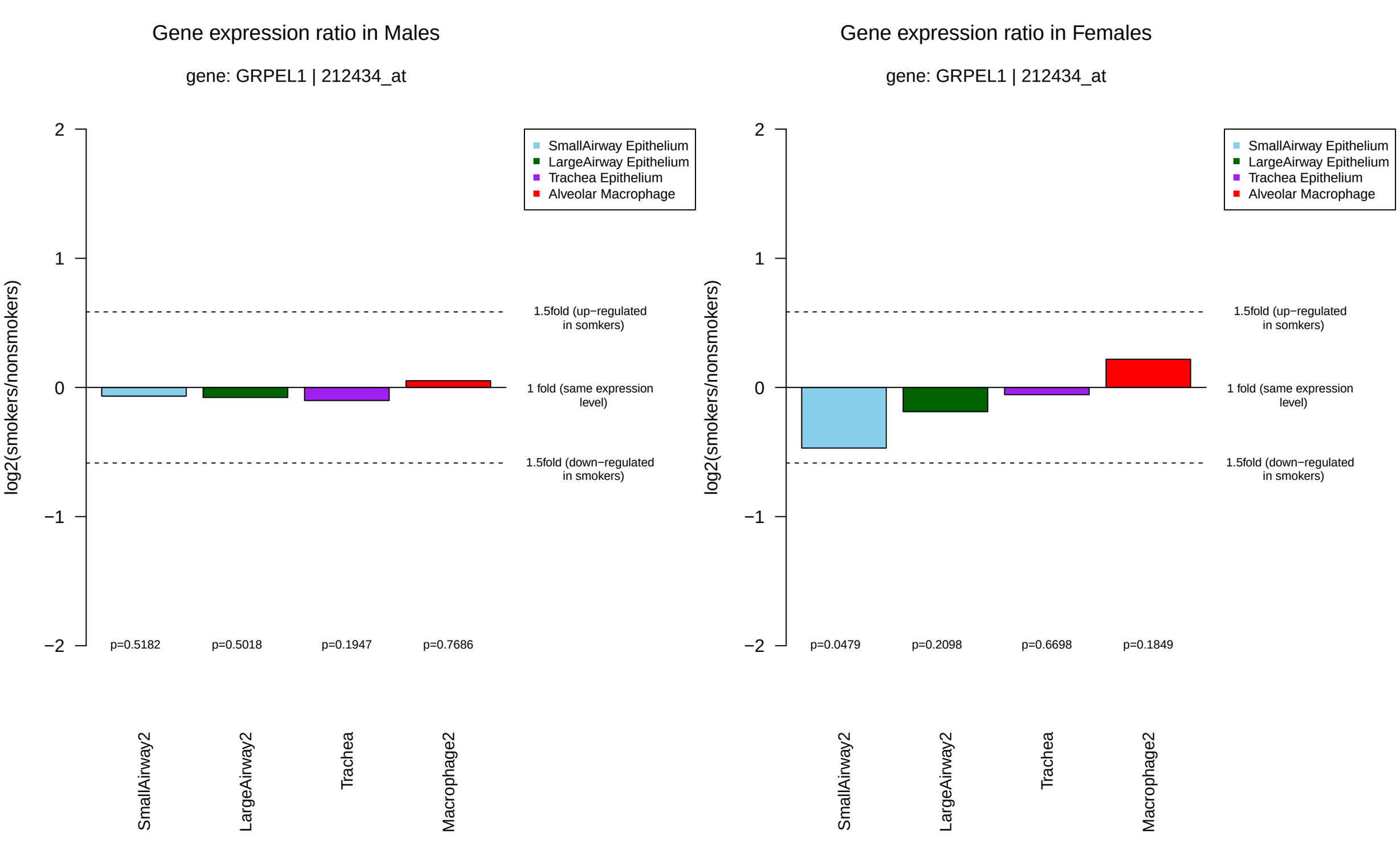
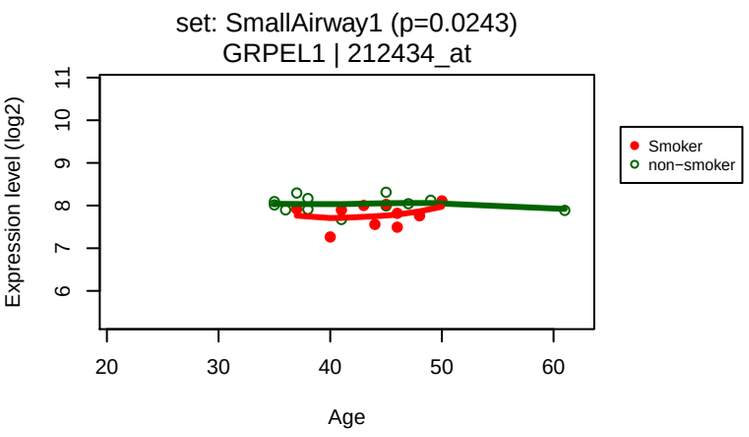
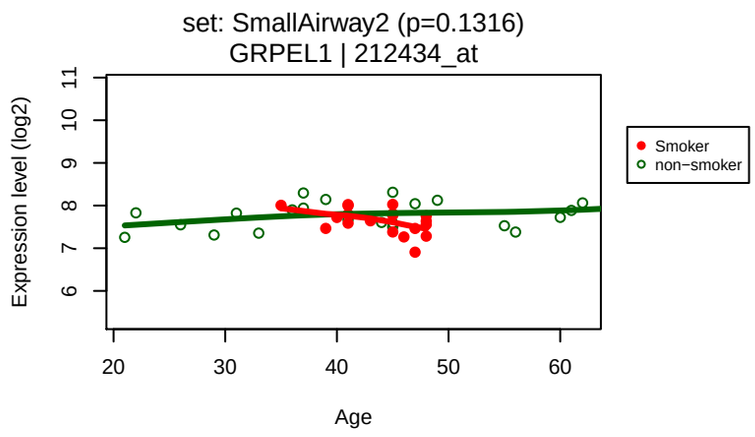
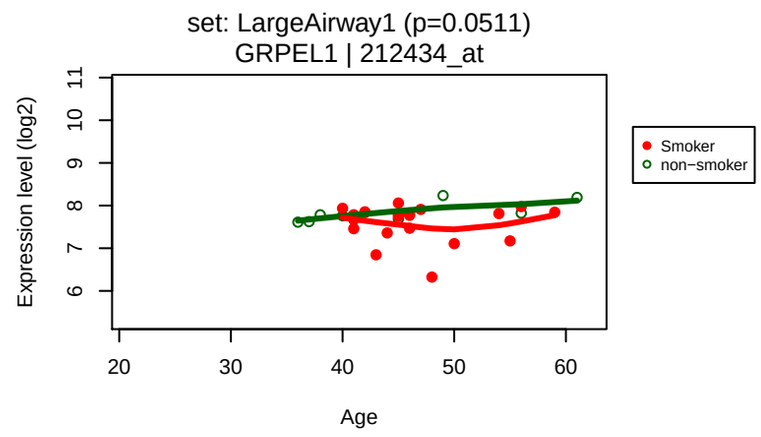

Gene expression of smokers and nonsmokers
gene: GRPEL1 | 212434_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0243
p=0.1316
p=0.0118
p=0.0511
p=0.1509
p=0.2542
p=0.021
p=0.4461
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

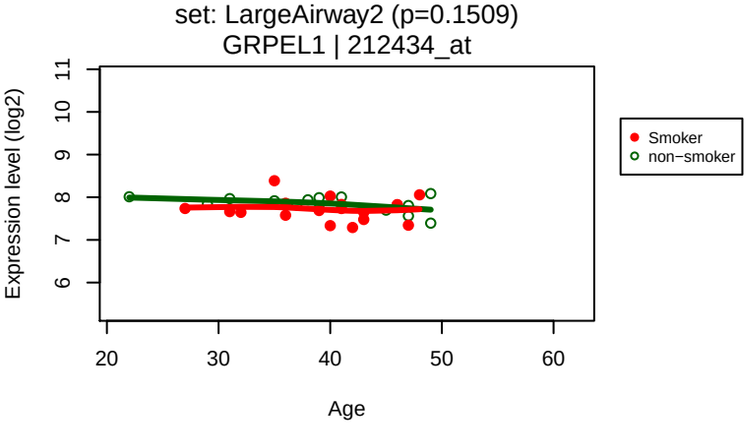

Gene expression ratio (smokers/nonsmokers)
gene: GRPEL1 | 212434_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0243
p=0.1316
p=0.0118
p=0.0511
p=0.1509
p=0.2542
p=0.021
p=0.4461
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

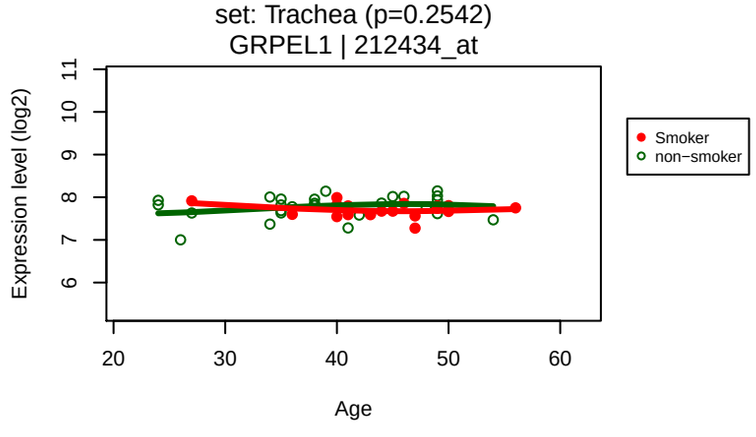

Gene expression ratio in Males
gene: GRPEL1 | 212434_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5182
p=0.5018
p=0.1947
p=0.7686
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRPEL1 | 212434_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0479
p=0.2098
p=0.6698
p=0.1849
SmallAirway2
LargeAirway2
Trachea
Macrophage2

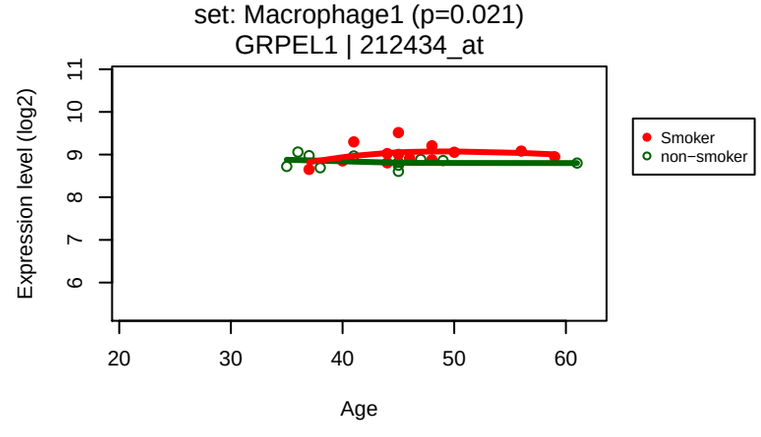

set: SmallAirway1 (p=0.0243)
GRPEL1 | 212434_at
Expression level (log2)
Age
Smoker
non−smoker

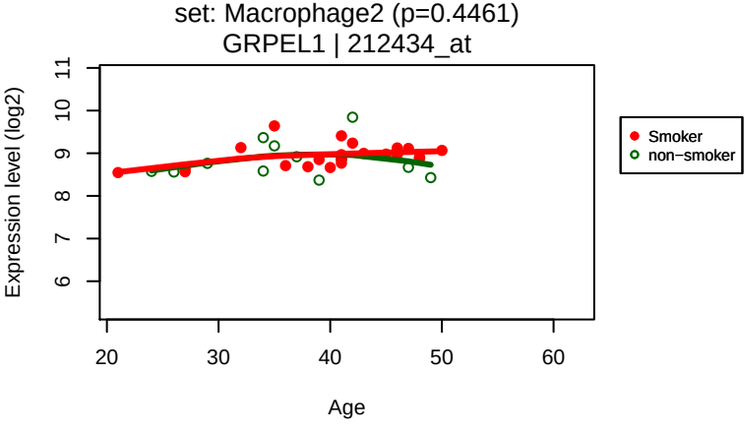

set: SmallAirway2 (p=0.1316)
GRPEL1 | 212434_at
Expression level (log2)
Age
Smoker
non−smoker

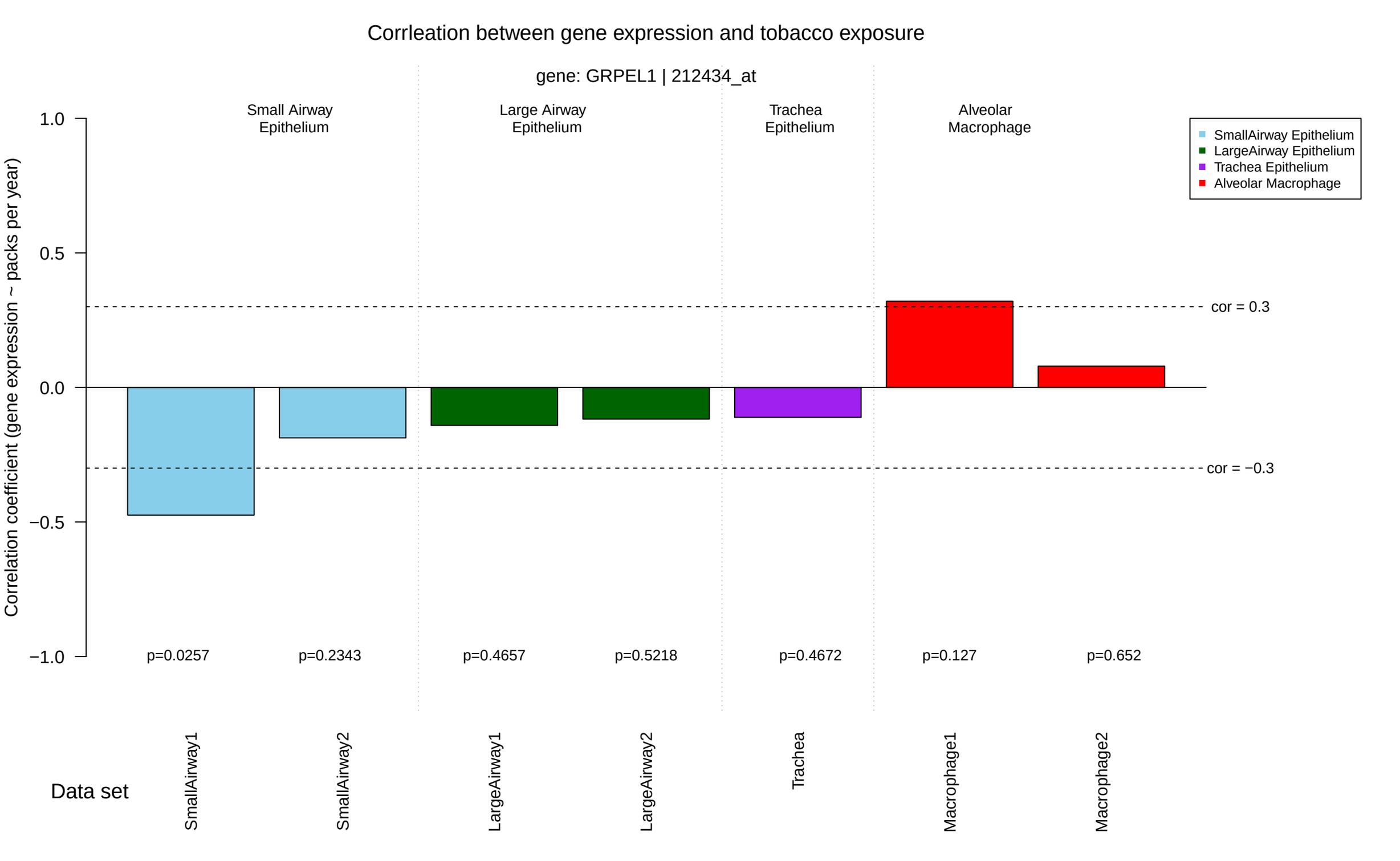

set: LargeAirway1 (p=0.0511)
GRPEL1 | 212434_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1509)
GRPEL1 | 212434_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2542)
GRPEL1 | 212434_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.021)
GRPEL1 | 212434_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4461)
GRPEL1 | 212434_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRPEL1 | 212434_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0257
p=0.2343
p=0.4657
p=0.5218
p=0.4672
p=0.127
p=0.652
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2